

Example 1: Quiz Score Claim

****Problem.**** Instructor claims average quiz score is 85.

Twelve quizzes yield $\bar{x} = 81.6$, $s = 6.3$. Test $\alpha = 0.05$ two-tailed.

****Solution.****

- $H_0: \mu = 85$, $H_1: \mu \neq 85$.
- $df = 11$, $t^* = \pm 2.201$.
- $t = (81.6 - 85)/(6.3/\sqrt{12}) \approx -1.86$.
- $p\text{-value} \approx 0.090 (> 0.05) \Rightarrow$ fail to reject H_0 .
- 95% CI: $81.6 \pm 2.201 \cdot 6.3/\sqrt{12} \Rightarrow (77.6, 85.6)$.
- Effect size $d = (81.6 - 85)/6.3 = -0.54$ (medium).
- Conclusion: Evidence insufficient to show the mean differs

from 85.

Example 2: Coffee Temperature Safety

****Problem.**** Safety spec says mean dispensing temp should be

140 °F. A shop samples 15 cups: $\bar{x} = 142.3$ °F, $s = 3.5$ °F. Test right-tailed ($H_1: \mu > 140$) at $\alpha = 0.01$.

****Solution.****

- $df = 14$; $t^* = 2.624$.
- $t = (142.3 - 140)/(3.5/\sqrt{15}) \approx 2.89$.
- $p \approx 0.006 < 0.01 \Rightarrow$ reject H_0 .
- 99% CI (two-tailed for reporting): $142.3 \pm 2.977 \cdot 3.5/\sqrt{15} \Rightarrow$

(139.6 , 145.0) — partially above 140.

- $d = (142.3 - 140)/3.5 \approx 0.66$ (medium-large).
- Conclusion: Coffee is significantly hotter than 140°F;

adjust machine.